

Skills Summary

Radical Equations and Extraneous Roots

Use this reference while completing your worksheet or when studying.

Skill 1 — Isolate the radical, then undo it

Method: get the radical alone on one side, then square both sides.

$$\sqrt{A} = B \implies A = B^2 \quad (\text{only usable when } B \geq 0)$$

Squaring before isolating leaves a cross term and makes the equation worse, so the isolating step always comes first.

Example: Solve $3\sqrt{x+2} - 5 = 7$.

Step 1. The radical is buried under a multiplication and a subtraction, so undo those first — squaring is only useful once the root stands alone.

Step 2. $3\sqrt{x+2} = 12$, so $\sqrt{x+2} = 4$, so $x+2 = 16$. Check: $3\sqrt{16} - 5 = 12 - 5 = 7$.
 $\implies x = 14$

Try it: Solve $4\sqrt{x-3} + 1 = 13$.

check: $x = 12$

Skill 2 — Test every candidate in the original

Why checking is not optional: squaring loses the sign, because 3^2 and $(-3)^2$ are the same number. So the squared equation can have solutions the original never had — **extraneous roots**. Substitute every candidate back into the *original* equation and keep only the ones that work. A candidate that makes the isolated radical equal a negative quantity always fails.

Example: Solve $\sqrt{x+6} = x$.

Step 1. The right side is a variable, so it could be negative for some candidates — that is exactly the situation where squaring invents an extra root and the check has to catch it.

Step 2. $x+6 = x^2$ gives $x^2 - x - 6 = 0$, so $(x-3)(x+2) = 0$ and the candidates are 3 and -2. Testing: $\sqrt{9} = 3$ works, but $\sqrt{4} = 2$ is not -2.

$\implies x = 3$ (the candidate -2 is extraneous)

Try it: Solve $\sqrt{3x+4} = x$, keeping only what survives the check.

check: $x = 4$ (the candidate -1 is extraneous)

Skill 3 — Solving a radical model for its input

Method: a model such as $v = k\sqrt{h}$ is solved the same way — divide by the constant, square, and state the answer in the model's unit. Only the non-negative solution can be a physical measurement, so the check also throws out negative candidates.

Example: A shallow-water wave travels at $v = 3\sqrt{h}$ metres per second, where h is the depth in metres. Find the depth giving $v = 12$ metres per second.

Step 1. The unknown sits under the root, so isolate the root by dividing by the constant and then square to release it.

Step 2. $3\sqrt{h} = 12$ gives $\sqrt{h} = 4$, and squaring gives $h = 16$. Check: $3\sqrt{16} = 12$.

$\Rightarrow$ **$h = 16$** metres

Try it: For $v = 5\sqrt{h}$ metres per second, find the depth giving $v = 35$ metres per second.

check: $5\sqrt{49} = 35$ metres